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Module :Free Software and Open Sour

Mester 1 Pharmaceutical Sciences

Theme :

Practical Work: Free and Open Source Software

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PART 1 – Theoretical Study of a Tool

1- General presentation of the tool:

Biopython is an open-source library based on the Python programming language, designed for the analysis and processing of biological data in bioinformatics. It provides tools for biological sequence manipulation, access to databases, sequence alignments, and molecular structure analysis. Biopython is widely used in scientific research to automate analyses and develop reproducible pipelines. It represents a flexible and accessible tool for biologists and bioinformaticians. (Cock et al., 2009).

2- Main functionalities:

Biopython provides a wide range of core functionalities for bioinformatics analyses, such as reading, writing, and manipulating common biological sequence formats (FASTA, GenBank, etc.). It enables programmatic access to biological databases and offers modules for sequence alignment and phylogenetic analysis. The library also includes tools for handling macromolecular structures and integrating external tools into automated workflows. These features make Biopython a comprehensive framework for computational biology applications. (Biopython 1.83 released, 2024).

3- Technical aspects :

Biopython is an open-source Python library dedicated to bioinformatics and computational biology, developed and maintained by an international community (Cock et al., 2023).

It allows manipulation and analysis of biological data such as DNA, RNA, and protein sequences programmatically in Python (Cock et al., 2023).

Biopython supports reading, writing, and converting common biological file formats such as FASTA, GenBank, and PDB (Cock et al., 2023).

It provides tools for sequence alignments, phylogeny, and biological data visualization (Cock et al., 2023).

The Bio.Entrez module enables access to biological databases such as NCBI (Cock et al., 2023).

Specialized modules exist for 3D structures, evolutionary trees, motifs, and clustering (Cock et al., 2023).

Biopython integrates well with the Python scientific ecosystem, including NumPy and matplotlib (Cock et al., 2023).

It is easily installable via pip and compatible with modern Python 3 versions (Cock et al., 2023).

The library is widely used to automate bioinformatics analyses (Cock et al., 2023).

Its reusable code facilitates the development of reproducible biological pipelines in Python (Cock et al., 2023).

4- Strengths:

-Bipython is a mature and freely available opensource Python library for bioinformatics and computational biology. (Cock et al., 2009)

- It provides comprehensive support for biological sequence manipulation, including DNA, RNA, and protein operations. (Cock et al., 2009)

- It supports reading, writing, and converting major biological file formats such as FASTA, GenBank, PDB, and more. (Cock et al., 2009)

-Biopython includes tools for sequence alignments and phylogenetic analysis useful in evolutionary studies. (Cock et al., 2009)

-It offers access to external biological databases (like NCBI) through modules such as Bio.Entrez. (Cock et al., 2009)

-The library has modules for 3D structure parsing and analysis, including support for PDB/mmCIF formats. (Cock et al., 2009)

1- Limitations and weaknesses :

Performance issues on very large datasets: Some operations (e.g., handling large genomes or high-throughput sequencing data) can be slow compared to specialized C/C++ tools. (Cock et al., 2009)

Limited advanced GUI or visualization tools**:** While basic plotting is supported, more sophisticated visualizations often require external libraries. (Biopython.org, 2025)

Dependency on external Python libraries: Certain modules require NumPy, SciPy, matplotlib, or ReportLab, which adds complexity to setup. (Cock et al., 2009)

Steep learning curve for beginners**:** Users need a solid understanding of Python and bioinformatics concepts to fully exploit the library. (Bio.Dev, 2025)

Partial coverage of some advanced bioinformatics tasks: Tasks like large-scale genome assembly, high-performance alignment, or molecular simulations are better handled by specialized software. (Cock et al., 2009)

2- Conclusion:

Biopython is a powerful and flexible Python library that simplifies the analysis and manipulation of biological data.
It covers the core bioinformatics needs, including sequence handling, alignments, phylogeny, 3D structures, and database access**.**
Its modularity and integration with the Python scientific ecosystem make it practical for automating reproducible workflows.

Despite some limitation such as performance issues on very large datasets, reliance on external dependencies, and limited visualization tools it remains a standard in modern bioinformatics.

Ref:

-Cock, P. J. A., Antao, T., Chang, J. T., Chapman, B. A., Cox, C. J., Friedberg, I., … de Hoon, M. J. L. (2009). *Biopython: freely available Python tools for computational molecular biology and bioinformatics.* **Bioinformatics, 25**(11), 1422–1423.

- *Biopython 1.83 released.* (2024). Open Bioinformatics Foundation.

-Cock, P. J., Antao, T., Chang, J. T., Chapman, B. A., Cox, C. J., Dalke, A., Friedberg, I., Hamelryck, T., Kauff, F., Wilczynski, B., & de Hoon, M. J. (2023). *Biopython: Freely available Python tools for computational molecular biology and bioinformatics* (Version 1.80)

-Biopython.org. (2025). *Biopython documentation*.

-Bio.Dev. (2025). *Biopython overview.*

PART 2 – Practical Study: Exploration of Zenodo

Introduction :

Zenodo is an electronic platform for scientific archiving launched by CERN (European Organization for Nuclear Research) in 2013 to support the publication and exchange of scientific research openly and freely. It allows researchers to deposit, share, and archive various types of data, software, and scientific publications, assigning them a DOI to ensure their citability in the long term. It relies on free software and stores its data in the CERN Data Center, ensuring safe and efficient preservation . (1)

*** The Main Objectives of Using Zenodo:**

Zenodo aims to provide a secure and reliable environment for researchers to publish and share their scientific research. The main objectives of using Zenodo are:

Enable researchers to share, archive, and disseminate research outputs freely: Zenodo provides an open digital repository where researchers across all disciplines can upload and preserve research outputs such as publications, data, software, presentations, and more. (2)

Provide persistent and unique identifiers (DOIs) for deposited work : Every upload on Zenodo receives a Digital Object Identifier (DOI), making it citable, traceable, and reliably linkable within the scholarly record. (2)

Maintain long-term preservation and accessibility : Research outputs are stored securely at CERN's infrastructure, ensuring their long-term preservation, access, and discoverability. (2)

Support versioning of research materials : Zenodo allows researchers to upload updated versions of their work while keeping records of previous versions for transparency and citation continuity. (2)

Facilitate interoperability and compliance with open science practices : The platform supports the FAIR principles (Findable, Accessible, Interoperable, Reusable), encouraging data management that enhances reuse and integration into global research infrastructures. (2)

Offer flexible licensing and sharing options : Users can select from a range of licenses, including open licenses like Creative Commons, to define terms of reuse and sharing according to their needs. (2)

* **The types of content hosted by Zenodo include:**

Zenodo accepts all file types and integrates with GitHub for depositing and long-term preservation of GitHub files . (3)

- Research data.
- Scientific publications and posters.
- Software.
- Text files.
- Spreadsheets.
- Interactive resources.
- Audio files , Video file and Images .

* **Importance of Zenodo for open science and research in NLS :**

Zenodo serves as a valuable resource for institutions and funding agencies seeking to comply with Open Science mandates. By providing a centralized platform for depositing research outputs, Zenodo simplifies the process of archiving and sharing, ultimately fostering a culture of openness and collaboration within the scholarly community. and facilitating the sharing of research materials and promoting transparency and rigor in scientific research, Zenodo plays a crucial role in addressing the reproducibility crisis.(4)

2 - Description of the steps carried out :

: **Search Performed 2.1**

. I accessed the Zenodo website (https://zenodo.org/)

I entered the following queryin the search bar : Endothelial cell

I chose this term because it contains the keyword “cell”, which is one of the required keywords in the assignment (cell, tissue, genome). It is also related to biological and medical .research

:After getting the results, I applied the filter

Resource type → Dataset

That's lead to find only datasets and exclude journal articlesor posters ....

2.2 : Dataset Selection Criteria

To selected dataset it's must include :the following criteria

Type: Dataset *

*Have open Access

Field: Natural Sciences *

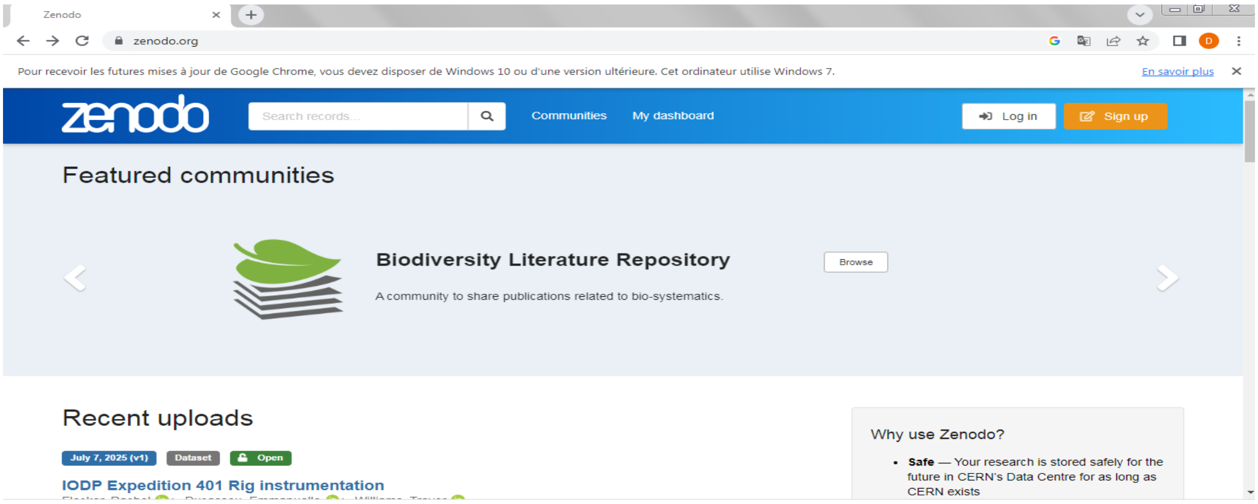
, Complete *structured . metadataand the presence of a DOI

) Recent publication *2023).

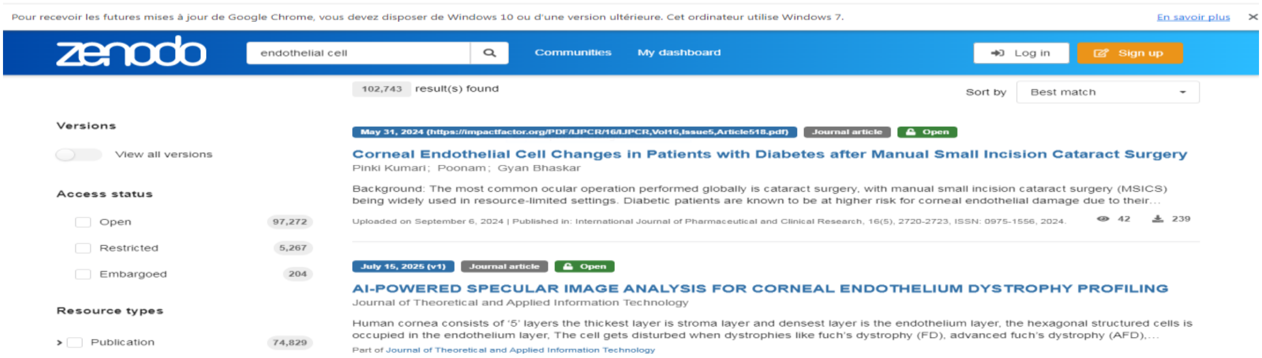
2.3 Navigation on the Platform

We've followed the following steps :

.1First thing, we accessed into Zenodo website: https://zenodo.org



.2Second , We entered "Endothelial cell" in the search bar.



. 3Third , we have a pplying the "Dataset" filterto avoid the journal scientific and the poster

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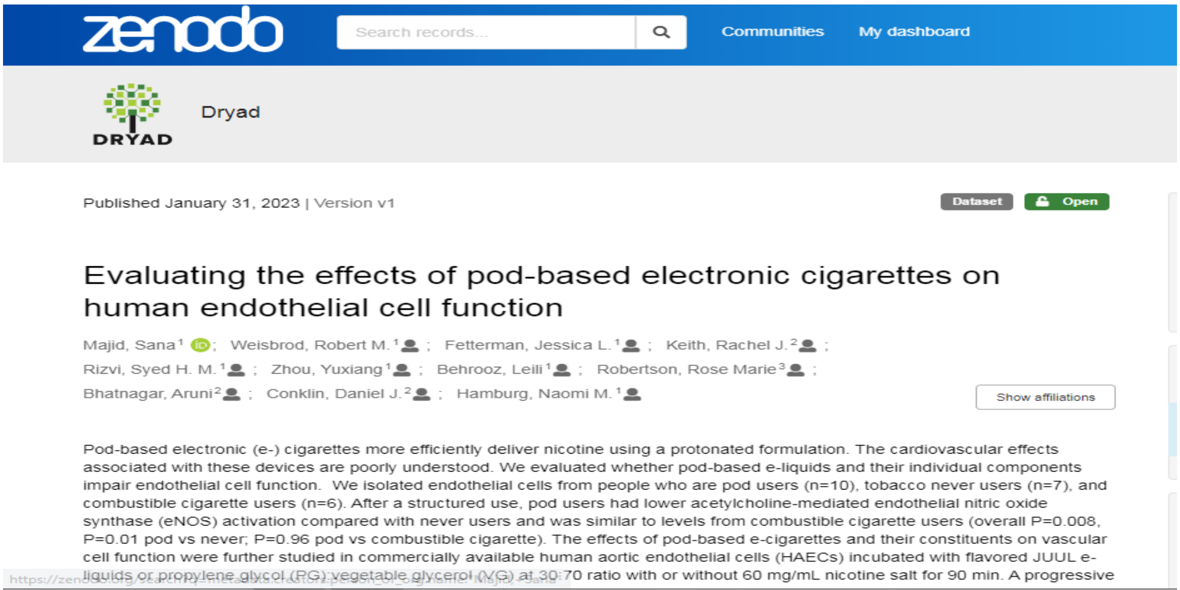
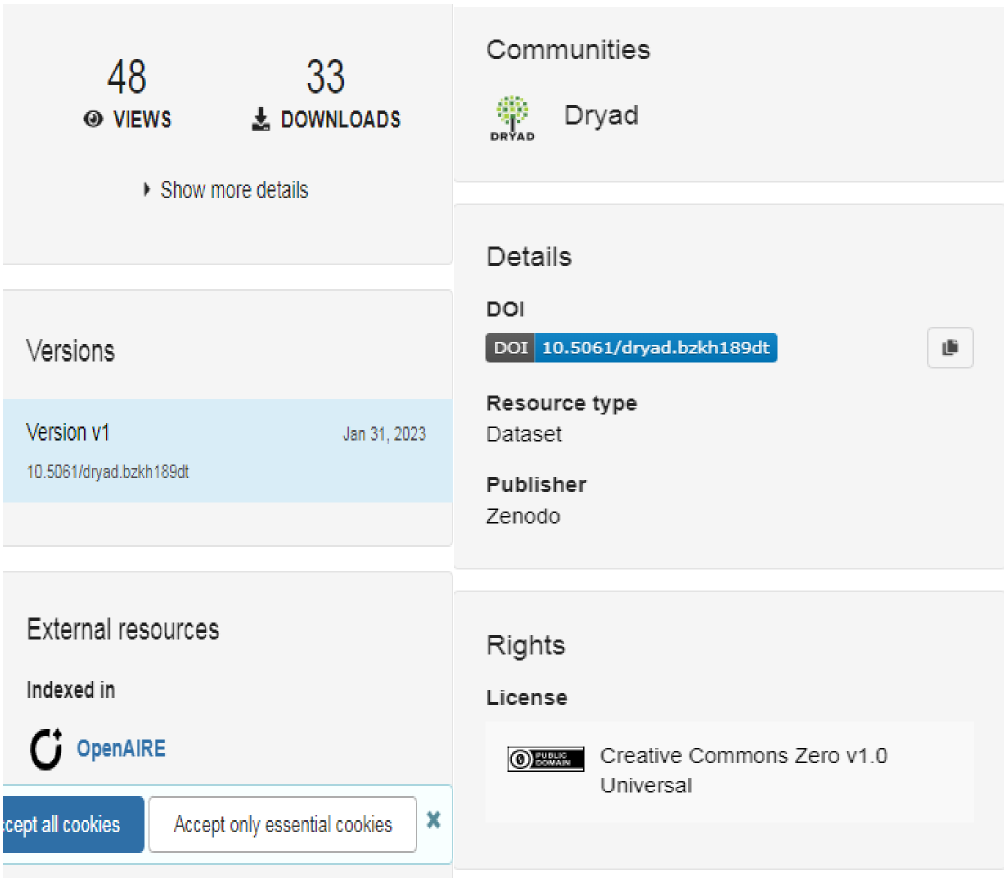
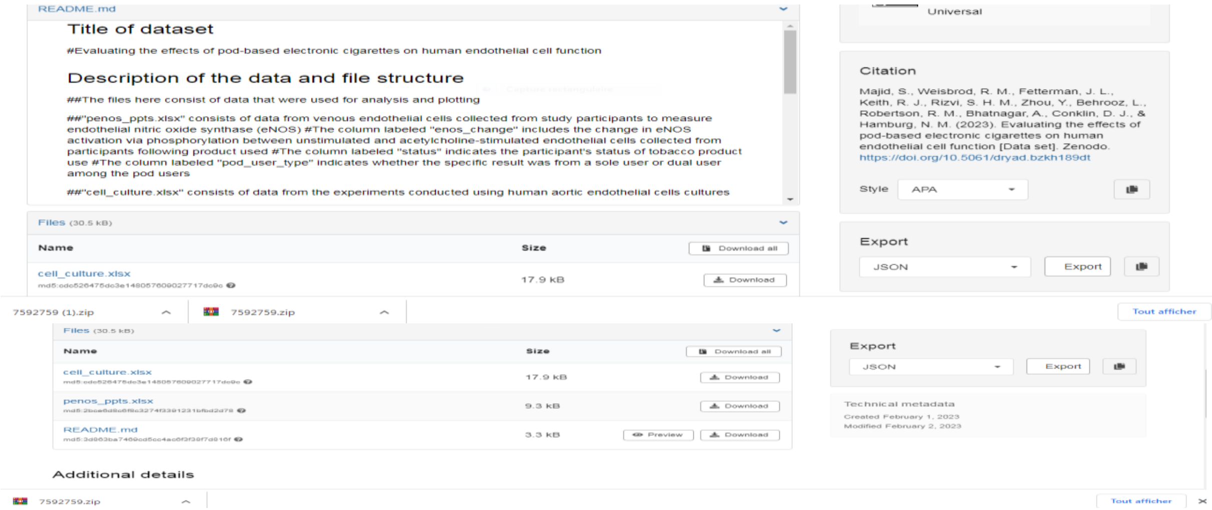
.4Fourth , Selected a relevant open-access dataset, we choose (Evaluating the effects of pod-based electronic cigarettes on human endothelial cell function).



.5Fifths , Opened the dataset page and reviewed the metadata.



.6Sixth, Downloaded the dataset (Version v1).



7. Seventh , ee have to do the extraction of the Metadata , according to Dublin Core standards.

3. Dataset Metadata :

The metadata were retrieved using the Dublin Core standard. We chose Dublin Core because it is an internationally recognized standard for describing digital resources, providing a simple and comprehensive framework for data description, which facilitates search, use, and exchange across different systems and platforms.

Element of Dublin Core	Information
Title	**Evaluating the effects of pod-based electronic cigarettes on human endothelial cell function**
Creature	**Majid, S.; Weisbrod, R. M.; Fetterman, J. L.; Keith, R. J.; Rizvi, S. H. M.; Zhou, Y.; Behrooz, L.; Robertson, R. M.; Bhatnagar, A.; Conklin, D. J.; Hamburg, N. M.**
Subject	**Human endothelial cells; e-cigarettes; cardiovascular function; cell biology**
Description	**Dataset containing measurements on the**

	function of human endothelial cells exposed to pod-based electronic cigarettes.
Publisher	**Zenodo**
Date	**January 1, 2023**
Type	**Dataset**
Format	**CSV / XLSX / Raw data (selon Zenodo)**
Identifier (DOI)	**https://doi.org/10.5061/dryad.bzkh189dt**
Source	**Dryad**
Language	**English**
Relation	**Dataset related to research on the effects of e-cigarettes on endothelial cells**
Rights (Licence)	**Creative Commons Zero v1.0 Universal (CC0 – Open Access)**
Coverage	**Human cell biology / life sciences**

6. Maximum Extraction of Available Information :

_ Technical metadata : That's include :

Version: v1

Views: 48

Downloads: 33

Data volume :352 .9 KB

Indexed in: OpenAIRE

Export : JSON

Additional details: it's cited by 10.1371/journal.pone.0280674 (DOI)

_ Citation (format APA) :

Majid, S., Weisbrod, R. M., Fetterman, J. L., Keith, R. J., Rizvi, S. H. M., Zhou, Y., Behrooz, L., Robertson, R. M., Bhatnagar, A., Conklin, D. J., & Hamburg, N. M. (2023). Evaluating the effects of pod-based electronic cigarettes on human endothelial cell function [Data set]. Zenodo. https://doi.org/10.5061/dryad.bzkh189dt

: Conclusion

We conclude that exploring Zenodo allowed us to find a relevant, well-documented, and easily accessible dataset in the field of cell biology. Zenodo provides a reliable open-access platform that facilitates the discovery, sharing, and reuse of scientific data. Thanks to the use of standardized metadata such as Dublin Core and Darwin Core, Zenodo ensures effective organization, interoperability, and transparency of research data, making it an essential tool for open science and the effective dissemination of scientific data.

Ref :

(1) : https://cernandsocietyfoundation.cern/projects/zenodo

(2): Wilkinson MD, Dumontier M, Aalbersberg IJ, Appleton G, Axton M, Baak A, Blomberg N, Boiten J, Da Silva Santos LOB, Bourne PE, Bouwman J, Brookes AJ, Clark TW, Crosas M, Dillo I, Dumon O, Edmunds S, Evelo CT, Finkers R et al (2016) The FAIR guiding principles for scientific data management and stewardship. Sci Data 3(1). https://doi.org/10.1038/sdata.2016.1

(3):https://datamanagement.hms.harvard.edu/share-publish/data-repositories/zenodo

(4) : https://openscience.cern/node/448